

Exercise 6 - Deep Learning

Code

- Dataset: class `SRDataset` in `code/dataset.py`
- Basic model: class `BasicSRModel` in `code/sr_model.py`
- Residual model: class `ResSRModel` in `code/res_model.py`

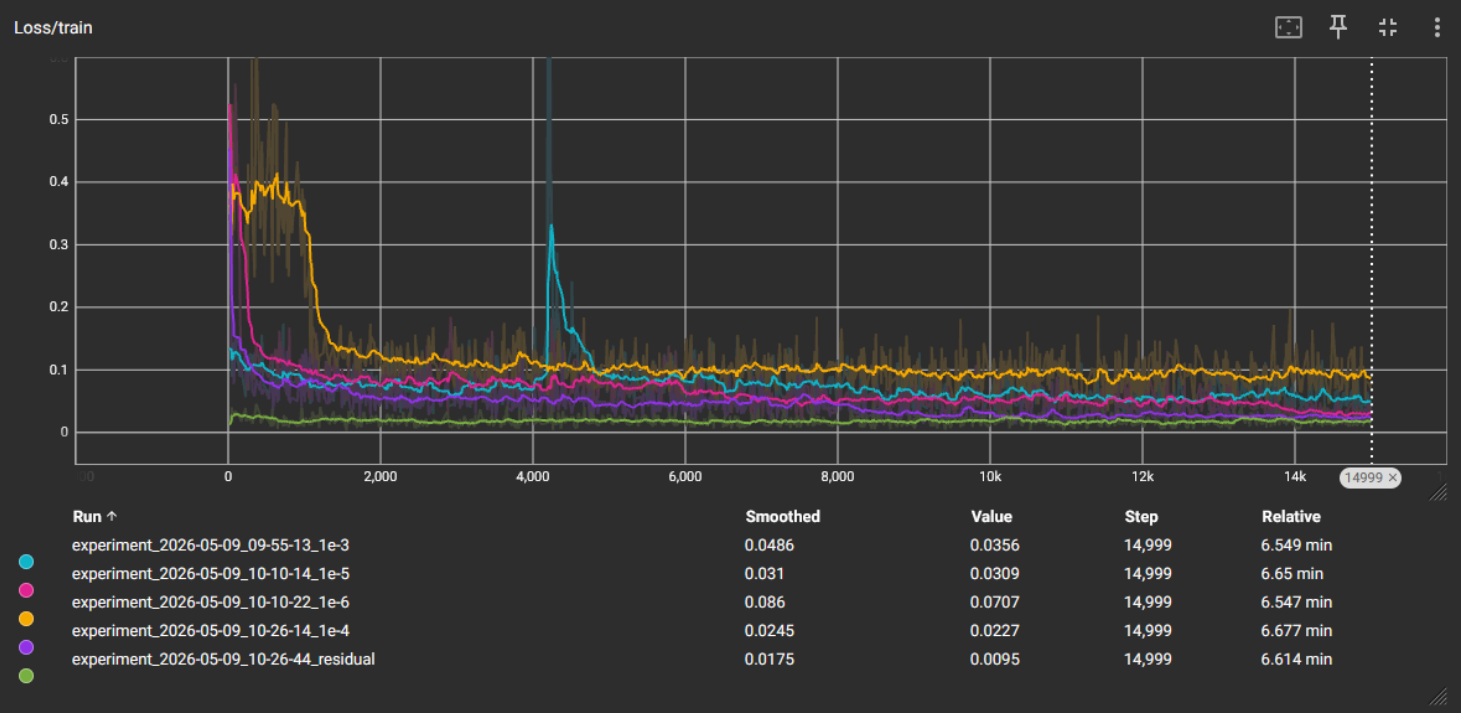
Executing both `sr_model.py` and `res_model.py` will print the number of parameters in the models.

- Training: file `code/train.py` . Can be given a `--model` argument which can either be `basic` or `residual` . This will start training and will write the model to disk every 50 epochs. The models are saved in `checkpoints/{run name}/model.pth` . The training process can be monitored with `tensorboard --logdir runs` .
- Evaluation: file `code/eval.py` . It will evaluate the model and print its PSNR and SSIM scores. It can be given two arguments to select the model:
 - `--run` : the name of the run folder in `./checkpoints/`
 - `--checkpoint` : the name of the model in that run. Defaults to `model.pth` , which is the final model after training.
 - Example: `python code/eval.py --run 2026-05-09_10-26-44 --checkpoint checkpoint_150.pth`
- An additional inference script in the file `code/inference.py` , which can be given the arguments `--run` and `--checkpoint` to select the model, and `--image` and `--output` . Example:
`python code/inference.py --run 2026-05-09_10-26-44 --checkpoint checkpoint_150.pth --image data/eval/0000.png --output upscaled.png`

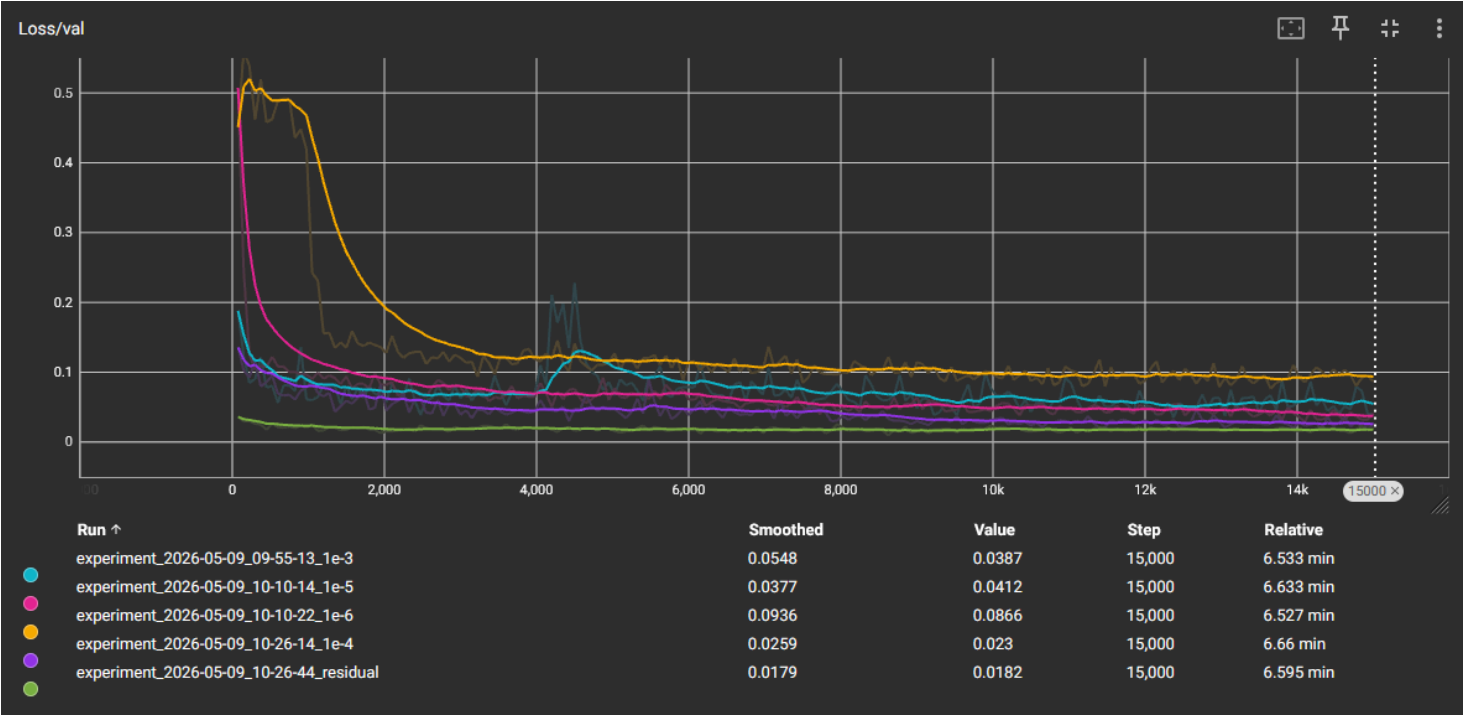
Models

Models are trained for 200 epochs.

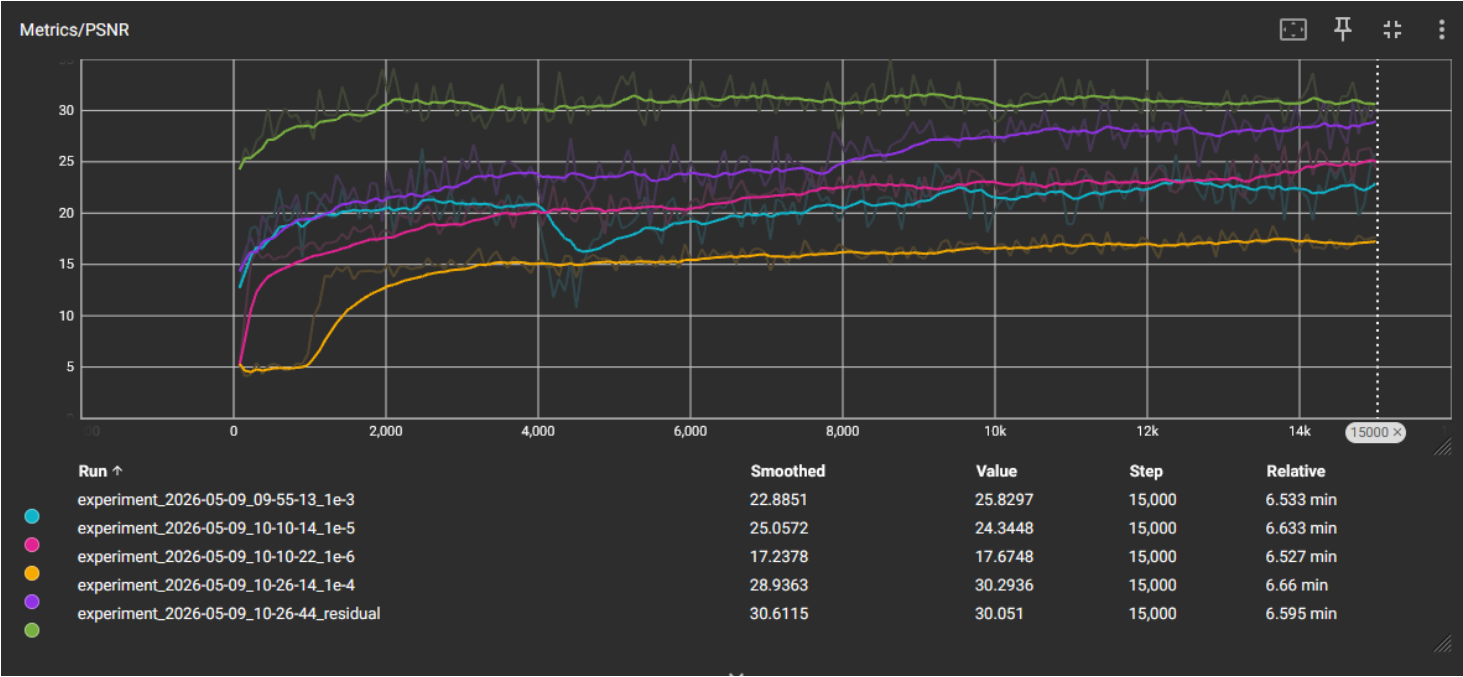
Train L1 loss (per iteration)



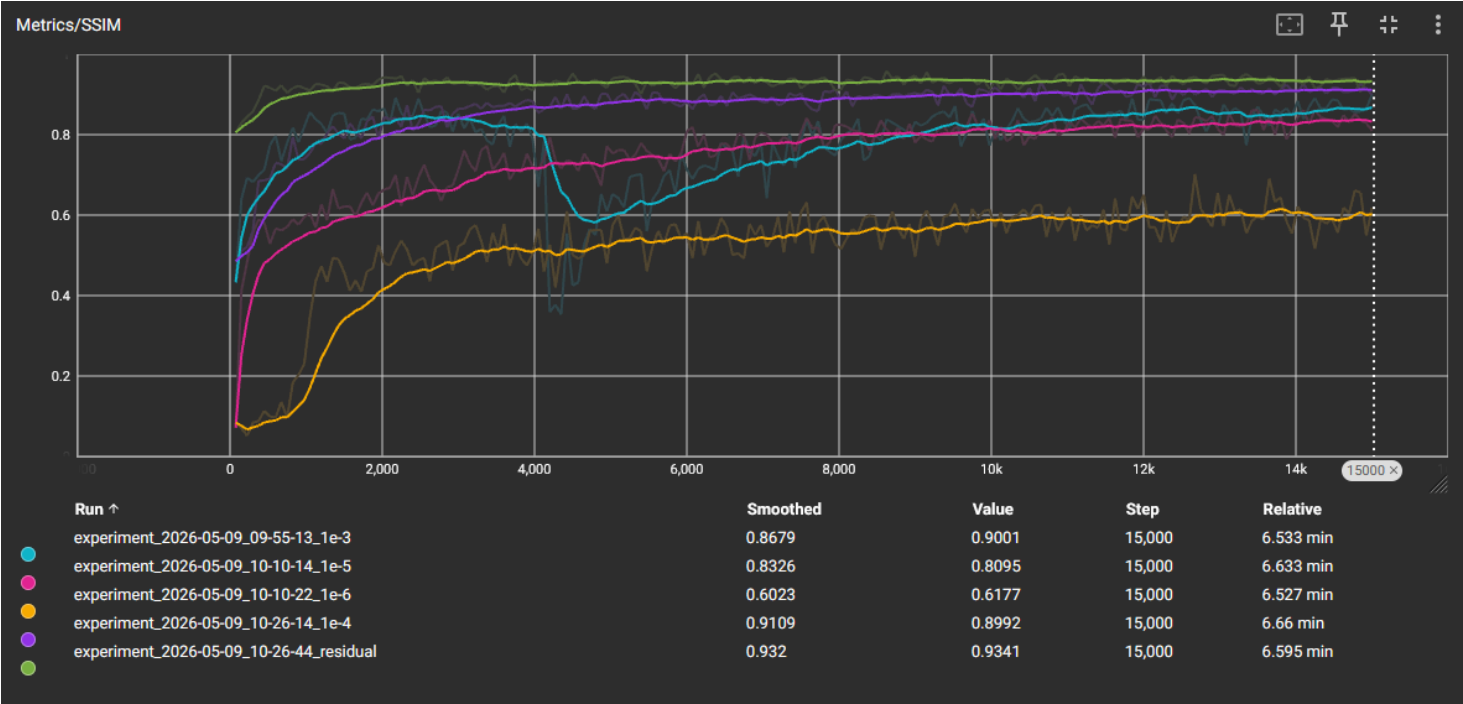
Validation L1 loss (per epoch)



PSNR (per epoch)



SSIM (per epoch)

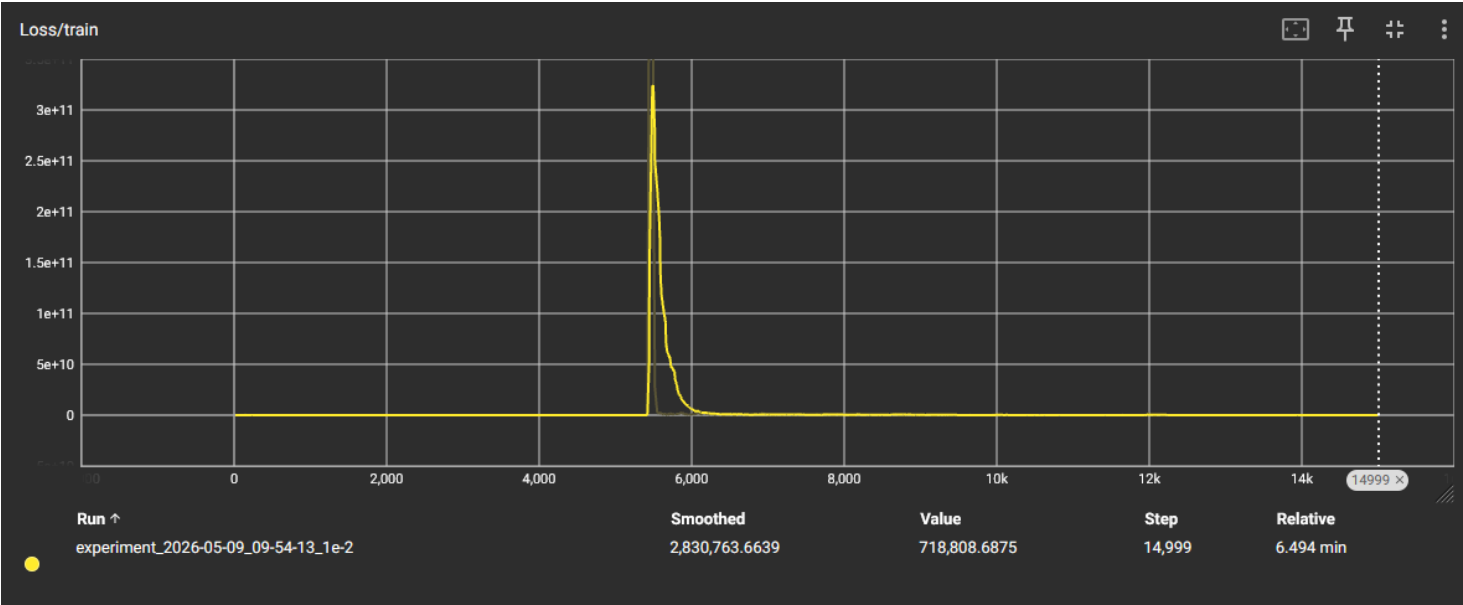


The above graphs only show the basic model with learning rates $1e-3$, $1e-4$, $1e-5$ and $1e-6$, and the residual models with learning rate $1e-4$.

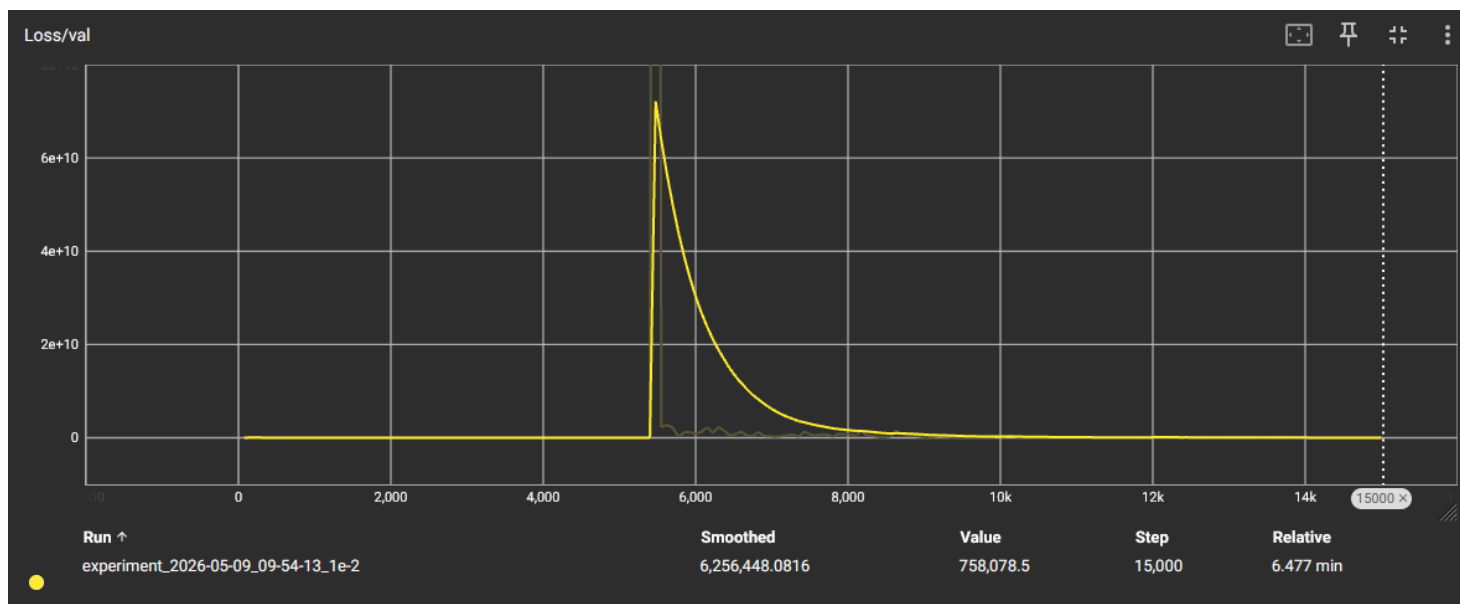
As can be seen, the residual model (with learning rate $1e-4$) outperforms all other models, and converges much faster. The basic model with learning rate $1e-4$ performs well and comes second across all metrics.

The basic model with learning rate $1e-2$ is omitted, as its performance is extremely poor and does not converge at all. Its metrics are shown below.

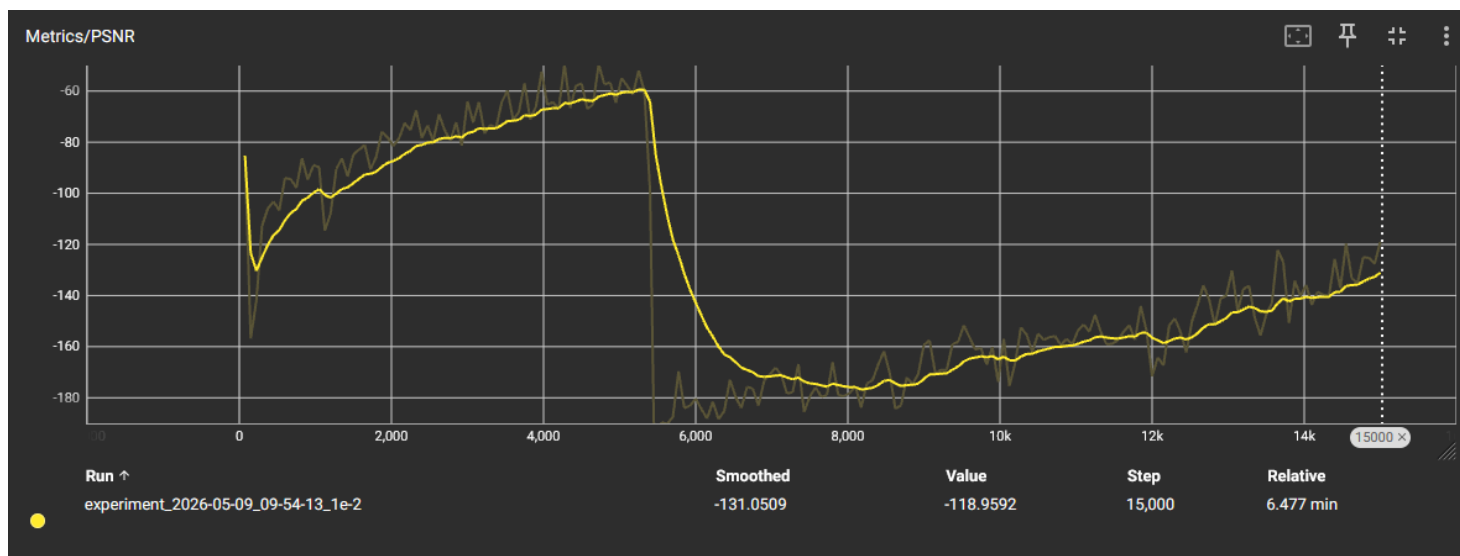
1e-2 model train loss



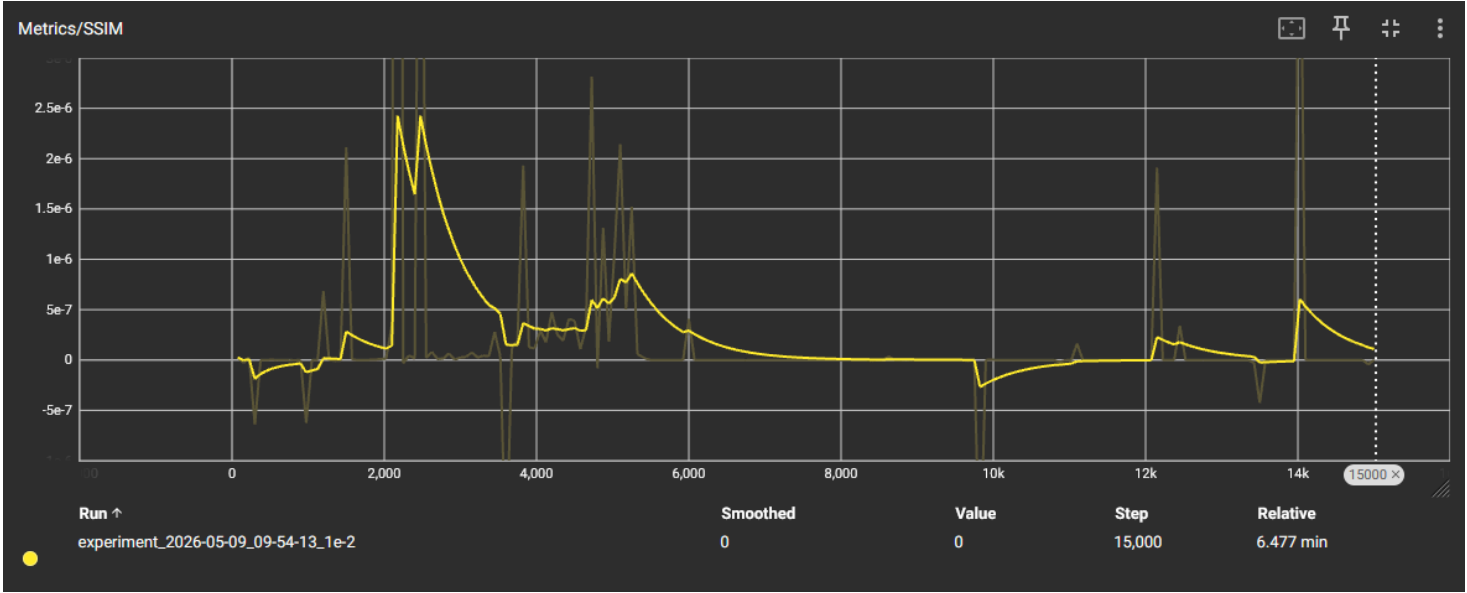
1e-2 model validation loss



1e-2 model PSNR



1e-2 model SSIM



The model completely diverges, with the training loss increasing to very high values and the validation loss, PSNR and SSIM are extremely low. This is expected, as a learning rate of 1e-2 is too high for this model and causes it to diverge.

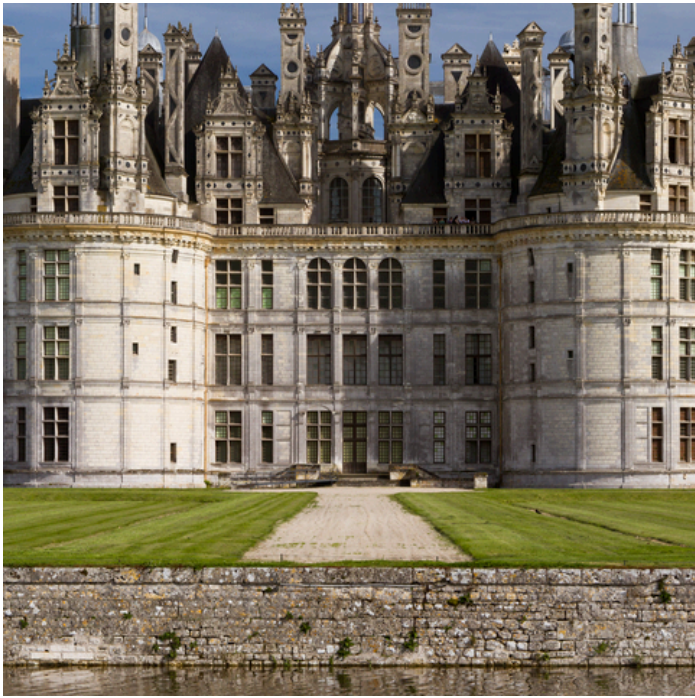
Numerical results

Models are sorted by PSNR (and equivalently by SSIM)

Model	PSNR	SSIM
Residual (lr = 1e-4)	28.26 dB	0.9154
Basic (lr = 1e-4)	27.27 dB	0.8865
Basic (lr = 1e-3)	25.16 dB	0.8697
Bicubic	25.07 dB	0.8390
Basic (lr = 1e-5)	24.49 dB	0.8234
Bilinear	23.82 dB	0.7874
Basic (lr = 1e-6)	17.46 dB	0.5557
Basic (lr = 1e-2)	-119.75 dB	0.0000

Visual results

Original castle image:



Upscaled image with the basic model (learning rate $1e-4$):



Upscaled image with the residual model (learning rate $1e-4$):

